What is Optimization Algorithm and its need?

- Based on knowledge gained from training data, machine learning models learn to generalize and make predictions about live data.
- The goal of machine learning models is to minimize the loss function or to close the gap between output data prediction and reality.
- The machine learning model will get more precise at predicting a result or classifying data as a result of iterative optimization.
- The iterative improvement for machine learning optimization of model configurations is known as hyperparameters.

Convex Optimization

- One of the most significant methods in the area of mathematical programming, which has numerous applications, is convex optimization.
- A convex optimization issue is an optimization problem where the goal is to identify a place using iterative calculations (usually, iterative linear programming) that maximizes or reduces the objective function.
- Both equality constraints and inequality constraints are applied to the objective function. An algorithm can be improved via convex optimization, which can hasten the rate at which it converges to the answer.
- Instead of computing an exact solution to the system, it can also be used to solve linear systems of equations.

Gradient Decent Algorithm

- Finding the minima of a function is done iteratively using the gradient descent method. This is a method of optimization for finding a function's lowest-valued parameters or coefficients. But this function can get stuck at a local minimum and does not always find a global minimum.

How does gradient descent work?

- It calculates the gradient.
- Gradient represents the change in loss with change in weights.
- The weights are modified according to the calculated gradient.
- These steps are repeated until we reach the minima of loss function.

Gradient Decent Algorithm

Limitations

- Gradient descent has the drawback that the weight update at a given time is solely determined by the learning rate and gradient at that time.
- The previous steps done when navigating the cost space are not taken into account.